

$$\begin{array}{ccc}
F(C, D) & \xrightarrow{F\left(\eta_C^{(D')}, D\right)} & F(U(D', F(C, D')), D) \\
\downarrow F(C, g) & & \downarrow F(U(g, F(C, D')), D) \\
F(C, D') & \xleftarrow[\epsilon_{F(C, D')}^{(D)}]{} & F(U(D, F(C, D')), D)
\end{array}$$